Exercise 3 - Backpropagation

05.05.2026

This exercise must be uploaded as a handwritten solution (scan or photo). As you will benefit from the upcoming lecture's content for solving this exercise, the **deadline is the 17th of May (two weeks)**. In total, the exercise gives 115 points. The percentages for each (sub)task are denoted in the title. Please leave us anonymous feedback for this exercise at https://survey.mbp.tf.uni-bielefeld.de/index.php?r=survey/index&sid=576629&lang=en.

Task 1: Gradients are Row Vectors (15%)

Consider a function $f : \mathbb{R}^{n\times 1} \to \mathbb{R}^{m\times 1}$, mapping n-dimensional column vectors to m-dimensional column vectors.

For any $\mathbf{x} \in \mathbb{R}^{n\times 1}$, the derivative $f'(\mathbf{x})$ is characterized by its linear approximation property:

$$f(\mathbf{x}+\mathbf{h}) \approx f(\mathbf{x}) + f'(\mathbf{x}) \cdot \mathbf{h}$$

for small $\mathbf{h} \in \mathbb{R}^{n\times 1}$.

Question:
Considering the dimensions of the involved entities $\mathbf{x}$, $\mathbf{h}$, $f(\mathbf{x})$, and $f'(\mathbf{x})$, explain why the gradient—i.e., the derivative of a scalar-valued function (when $m = 1$)—is a **row vector**.

Task 2: Gradient of a Bilinear Form (5% + 10% + 20%)

Let $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{y} \in \mathbb{R}^m$ be column vectors.

The bilinear form is defined as

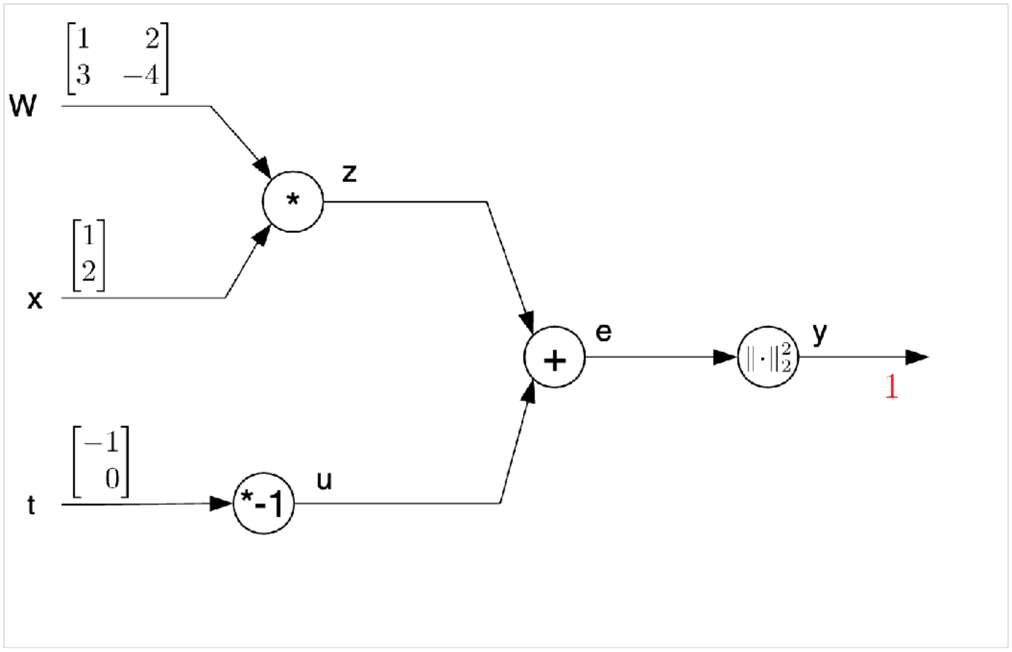
$$f(\mathbf{x}, \mathbf{y}, W) = \mathbf{x}^T W \mathbf{y} = \sum_i x_i \sum_j w_{ij} y_j$$

and yields a scalar output.

1. What is the correct dimension of the matrix W?
2. Determine the dimensions of the following derivatives:
   (a) $\nabla_{\mathbf{x}} f$ (standard row-vector gradient)
   (b) $\nabla_{\mathbf{x}^T} f \equiv \nabla_{\mathbf{x}}^T f$ (column-vector gradient)
   (c) $\nabla_{\mathbf{y}} f$ (row-vector gradient)
   (d) $\nabla_{\mathbf{y}^T} f \equiv \nabla_{\mathbf{y}}^T f$ (column-vector gradient)
3. Compute the expressions for each of the derivatives listed above.

Task 3: Computation Graph (5% + 10% + 15% + 20%)

Consider the following computation graph:



1. Write the computation graph as a formula: $y =$
2. Perform a forward pass using the given values for $\mathbf{W}$, $\mathbf{x}$, and $\mathbf{t}$.
   You can denote intermediate results directly in the graph above the connecting arrow lines if you printed this exercise.
3. Determine the local gradients of all operation nodes, in component-wise notation:

$$\begin{aligned} + : \quad & \frac{\partial e_i}{\partial z_j} = \delta_{ij}, \quad \frac{\partial e_i}{\partial u_j} = \delta_{ij} \\ \| \cdot \|^2 : \quad & \\ \times : \quad & \\ \times\text{-1} : \quad & \end{aligned}$$

4. Perform a full backward pass (backpropagation), computing the derivatives for all nodes. You can denote the results below the respective arrows (see red 1) if you printed this exercise.